

ALFAYEZ AHMAD

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SUMMARY

Systems Developer and AI & Robotics Researcher taking a "silicon-up" approach to engineering. Specialized in the intersection of autonomous navigation, embedded control, and stochastic optimization, ensuring high-level AI architectures are grounded in mathematical rigor and hardware efficiency.

EDUCATION

Integral University <i>B.Tech in Computer Science & Engineering (Cloud Computing & AI)</i> Cumulative GPA: 9.55 / 10.0 Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Advanced Java	Lucknow, Uttar Pradesh 2024 – Present
Council for the Indian School Certificate Examinations (CISCE) <i>The Indian School Certificate (ISC)</i>	Grade: 92.25%

PUBLICATIONS

Reinforcement Learning Controlled Variable Speed Limits in Urban Expressway Mixed Traffic <i>IEEE Xplore Digital Library (IMPACT 2026, Aligarh Muslim University)</i>	Feb 2026
- Engineered a Reinforcement Learning (RL) based Variable Speed Limit (VSL) control framework using a Deep Q-Network (DQN) implemented in the SUMO environment. - Achieved a documented 18.8% reduction in total travel time compared to baseline, effectively mitigating congestion and shockwave formation.	
Foundations of AI-Driven Precision Medicine: Algorithms, Omics, and Clinical Integration <i>Chapter 10, Bentham Science: Artificial Intelligence for Next-Gen Healthcare</i>	Jan 2026
Authored a specialized chapter focusing on the technical integration of AI algorithms with multi-omics data to advance diagnostic accuracy.	

EXPERIENCE

Lead Developer & Researcher (Open Source) <i>Independent Research</i>	Nov 2025 – Present
- Architected SentinelA, a real-time Network Intrusion Detection System (NIDS) optimized for Apple Silicon (macOS ARM64) utilizing Scapy for promiscuous mode packet analysis. - Deployed Random Forest machine learning classifiers for live traffic inference, achieving sub-millisecond latency in detecting high-velocity Port Scans and DoS attacks. - Engineered asynchronous Python data pipelines integrated with SQLite to capture edge-case network telemetry and enable high-throughput performance logging.	
Technical Community Contributor (Drones & Robotics) <i>Hackster.io</i>	2019 – 2023
- Developed and documented flight control firmware for quadcopters using STM32 (ARM Cortex-M) and Atmega328P (AVR) systems. - Implemented Extended Kalman Filter (EKF) algorithms for UAV state estimation to fuse IMU, GPS, and barometer data. - Architected nested-loop PID controllers with 400Hz+ stabilization and managed I2C/DMA protocols for hardware-software interfacing.	

TECHNICAL PROJECTS

Autonomous Warehouse Perception System (SAM 2 + SQL)	Dec 2025
- Integrated Meta's Segment Anything Model 2 (SAM 2) with OpenCV morphological operations to autonomously track objects undergoing severe topological changes and fragmentation. - Engineered a custom "Multi-Centroid" visual prompting strategy, ensuring robust ID persistence across complex 1-to-4 object split scenarios in dynamic environments.	
Quadcopter-Sim-V1: Autonomous Flight Pipeline & Digital Twin	Oct 2025
- Developed a modular 1D flight simulator mimicking industrial robotics stacks with a discrete-time Newtonian physics engine using Euler Integration. - Implemented a Linear Kalman Filter to suppress Gaussian sensor noise and engineered a PID feedback loop with gravity feed-forward compensation.	
PM2.5 Air Quality Forecasting: Cloud-Ready ML Microservice	Aug 2025
- Engineered a Docker-containerized ML pipeline to forecast daily PM2.5 concentrations based on localized meteorological data. - Conducted comparative analysis between Linear Regression and a Random Forest Regressor, capturing sharp non-linear pollution peaks.	

TECHNICAL SKILLS & CERTIFICATIONS

Languages: C, Java (Advanced), Python (Advanced), SQL (SQLite). AI/ML: TensorFlow, Keras, Scikit-Learn, Reinforcement Learning (MDPs, DQN, PPO). Robotics & Vision: ROS 2, PX4, OpenCV, TensorRT. Hardware & Deployment: STM32 (ARM), Atmega328P, Apple Silicon, GNU/Linux, Docker, Git. Certifications: IBM Machine Learning with Python & Cloud Application Developer Google Cloud (Compute, Vertex AI, API Gateway).	
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